

Abhay Manhas

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PROFESSIONAL SUMMARY

Accomplished Python Developer with **3+ years** of in-depth knowledge. Expert in building robust web applications using Django and Rest Framework. Skilled in asynchronous programming and real-time solutions with websockets. Proficient in designing background processing system utilizing RabbitMQ and Celery. Transitioned into AI and ML technologies, mastering pytorch and core concepts like linear regression, classification, backpropagation, optimization and loss functions.

PROFESSIONAL EXPERIENCE

Senior Python Developer

April 2023 – Present

Shubpy Pvt. Ltd.

Chandigarh

- Architected a **hybrid-cloud AI orchestration pipeline** using OpenRouter to dynamically route jobs to Gemini, Azure OpenAI and Qwen, **reducing AI task costs by 35%**.
- Engineered an **agentic RAG system utilizing MCP connectors** to trigger real-time operational workflows, achieving a 0.85 F1 accuracy score on retrieval and execution benchmarks.
- Built a MapReduce-style analysis service with **Celery** to process large documents, enabling summarization of **500+ page texts in under 1 minute**.
- Deployed appliaton on **Aks(Azure kuberetes services)**,Auto scaled the system to handle high traffic and Implemented **CI/CD pipeline in Azure DevOps**, achieving unit test coverage and **reducing deployment incidents by 90%**.

Junior Python developer.

June 2022 - Dec 2022

CNT Technologies

Chandigarh

- Built real-time chat application using **asynchronous programming and WebSockets**,enabling instant communication for 100+ concurrent users with message **latency under 100ms**.
- Designed Multi-queue system withm **RabbitMQ Cluster** ensuring fault-tolerant communication between services.
- Migrated legacy **monolithic** Python applications into **Microservices architecture**.Turnd messy code into Three layer archiecture while maintaining code boundaries.

PROJECTS

AILYZE (Ai Qualitative Analysis Tool)

www.ailyze.com

- Developed the backend into a **modular Clean Architecture**, decoupling business logic from data access layers. This enforced strict separation of concerns, reducing debugging cycles and maintenance **overhead by 60%**.

Food Ordering Website

www.chds.com.au

- Engineered and deployed a consolidated high-performance backend capable of sustaining **10K+** concurrent requests with minimal latency. Optimized database connection pooling to handle peak traffic loads without degradation.

TECHNICAL SKILLS

Languages & Frameworks :Python, Django, Django REST Framework (DRF), Channels, WebSockets, Redis, Asyncio, Celery, Pandas, Numpy.

DevOps & Cloud Infra :AWS, GCP, Azure, Docker, Azure Kubernetes Service (AKS), CI/CD, Git, Azure Functions, Blob Storage, Azure Database for PostgreSQL.

AI/ML System & Integrations :PyTorch, Scikit-learn, Hugging Face Transformers, Model Training & Evaluation, Crewai, Gemini, OpenAI, OpenRouter, Claude, RAG, NLP, MCP Server.

Tools& Platforms :Zapier, n8n, Prometheus, Grafana, Flower, Pytest, Unittest, Stripe.

CERTIFICATIONS & ACHIEVEMENTS

- Completed 25-hour PyTorch Deep Learning Course | [Link](#)
- CrewAI Certified - Multi-agent artificial intelligence systems | [Link](#)
- Python Certified, HackerRank | [Link](#)
- KCNA — Kubernetes & Cloud Native Associate | [Link](#)

EDUCATION

Gura Nanak dev university

Bachelor Of Computer Application

Amritsar

Aug. 2019 – May 2022